

Appendix A - First Interview Transcript

Transcript - first interview (The Scenario)

Client name : [REDACTED]

Contact details : +91 8105933111

Date of interview : 16 Feb 2025

Place of interview : Client's house

Hello [REDACTED], thank you for taking the time for this interview. Today, we will be establishing the background and requirements of your role as well as the problem you are currently facing.

Client: Alright.

Could you explain your job and briefly describe your business?

Client: I work for Microland Limited, a global IT infrastructure services company headquartered in Bangalore. We provide IT infrastructure services to clients worldwide and have around 4,500 employees across various locations. My role is Vice President of Human Resources, where I am responsible for HR partnership and operations. I lead a team of 35 people across different regions.

Could you explain how exactly the budget planning process works in your departments?

Client: Sure. In the last quarter of each year, every department must submit a request for the budget they need in the next fiscal year. This includes manpower costs, travel, miscellaneous fees, and costs for key programs or facilities to be implemented. The HR budget team then analyzes past trends to assess where departments have spent more or less than required and considers any new programs the departments may want to introduce. Based on this analysis, the budget planning team reviews and adjusts the requested budgets to ensure they align with both the company's financial strategy and the departments' operational needs. For example, if a department underspent its

allocated budget the previous year and has no new programs to implement, the budget team would consider giving it a reduced budget in the next fiscal year.

That makes sense. Given this structured process, could you elaborate on your specific responsibilities in the budget planning?

Client: My responsibilities include understanding the business plan, growth forecast, and aligning HR budget requirements accordingly. I own the tracking of the budget for the entire HR department, including monitoring planned versus actual expenses across all budget categories. This involves managing budgets for approximately 88 people across different locations, ensuring we remain within our allocated funds, and accurately forecasting our budget for the next fiscal year.

Can you describe the current budgeting system and how your team collects input from different regions?

Client: Currently, the finance team provides us with an Excel sheet every month, which outlines the planned versus actual spending on a month-to-month basis. This sheet is shared only with the HR leadership team.

I see. Given that the current budgeting process relies heavily on manual tracking, what challenges have you encountered with this system?

Client: The process of manually reviewing and distributing budget information to each individual department is extremely time-consuming. Since each department has an individual budget sheet, keeping track of all their expenses at once and aligning them to the overall business plan can be challenging and disorganised. Additionally, because of the manual nature of the process, we sometimes miss reviewing data every month. Errors, such as tagging expenses to the wrong department or data inconsistencies and a lack of data security, are also prevalent and must be manually corrected, which takes even more time. It is also difficult to regulate access to these confidential documents.

You mentioned that your departments consist of employees from different regions with different roles. How exactly do you track budgets for each individual department in a standard form, and in what way is the budget categorised? Is it region-wise or task wise?

Client: Each sub-function within HR—Talent Acquisition, Talent Management, Compensation & Benefits, Business HR, HR Operations, Internal Communications & Design, and Learning & Development—has different budgetary needs.

Budget requests are captured in an Excel sheet, where we provide year-to-date spend data for review. Each sub-function or department must analyze past trends, assess their needs for the upcoming fiscal year, and determine whether they need to add, drop, or adjust budget line items based on their priorities.

Once we finalise the budget, we send it to each individual department in the form of an Excel sheet so that they can track their expenses accordingly.

What are the key criteria or guidelines that teams must follow when requesting a budget?

Client : I'll list them out for you. Teams are required to:

- Review their year-to-date spend data.
- Analyze past budget trends.
- Determine new budget requirements based on priorities. (eg. if they wish to start a new program next year that will take up a significant chunk of their budget, is it worth the extra expense?)
- Justify the necessity of their requests, especially if they have underspent in the previous year.
- Ensure that all requests align with business objectives.

How do you communicate budget updates and approvals to different teams, and how many people have access to all budget data?

Client : Budget updates are shared via Excel sheets, and relevant cuts are communicated accordingly. Once budgets are frozen for the year, they are final, except in cases of significant business changes—such as losing key clients (which may require budget cuts) or securing major clients (which may allow increased spending to support business growth).

Only the heads of each department have access to the budget data. They submit budget requests, and they are informed of what has been approved.

To summarize, the current budget allocation and tracking system is manual, time-consuming, and susceptible to data discrepancies. You're looking for a centralized portal where departments can submit budget requests, access approved allocations, track planned versus actual spending in real time, and categorize expenses by department. Does this capture your key requirements?

Client : Yes, this sounds quite cohesive.

Perfect. Before we wrap up, are there any additional features you would like to see implemented?

Client: Yes, it would be helpful if departments could input their expenses in the portal as they occur, ensuring real-time tracking. These updates should be flagged to the budget controller (admin) for approval. This would help eliminate errors, improve transparency, and streamline the entire budgeting process.

Sure. Thank you for your time, Mrs. [REDACTED] If you think of any additional features that could enhance the solution, please let me know. I'll also reach out if I encounter any issues or need further clarification—would that be alright with you?"

Client : Sure, no problem.

Client Signature : Feb 18, 2025.

Usha.R

Appendix B - Final Interview Transcript

Transcript - Final interview (The Solution)

Client name : [REDACTED]

Contact details : +91 8105933111

Date of interview : 16 Feb 2025

Place of interview : Client's house

Hello Mrs. [REDACTED], I hope you're doing well. Thank you for taking the time to speak with me again. This interview is being conducted for my IB Computer Science coursework to evaluate the final solution — the budget tracking web application I developed for Microland. Could you tell me your overall impressions of the system?

Client: Good afternoon. Yes, I've had the chance to explore it quite a bit, and I'm genuinely very happy with how it turned out. It's clean, fast, and so much more organised compared to the Excel-based system we were using before. Everything feels centralised — I can log in, see all departments, track their budgets, and know exactly what's pending or approved. It's a big step up in both efficiency and security.

That's great to hear. Let's start with the login and authentication process. How did you find that?

Client: The login process worked smoothly. I liked that every user type — Admins, Department Heads, and Employees — has separate access. The role-based system makes sure that no one can view or edit something they're not supposed to. For example, department heads can only work with their department budgets, while I can see everything as the admin. That separation is really important for us, as we've had issues with data tampering in the past.

I'm glad that was effective. Did you try resetting your password or staying logged in across pages?

Client: Yes, both. The password reset feature worked perfectly — it's quick and sends an email immediately. I also liked that once I logged in, I could move between pages without having to log in again every time. That's very convenient. I remember you mentioning something about "session management" — I think that's what you meant, right? It definitely makes things easier while still keeping it secure.

Exactly. The system keeps you logged in while you're active but logs you out automatically after inactivity for security.

Client:: Yes, I noticed that! I stepped away for some time, and it automatically logged me out — that's very reassuring.

Let's talk about the admin functionalities. How did you find creating, editing, and managing user accounts?

Client: Very intuitive. I could easily create new users, assign them roles, or even delete old accounts when people left. It helps maintain control over who has access. The filters and search bar for departments and roles are really handy — I can quickly find specific employees without scrolling through a long list.

Great! And how about the budget management section — editing departmental requests or approving them?

Client: That worked exactly as expected. I could view all pending requests, approve or reject them, and the changes reflected immediately in the system. The ability to leave comments or feedback for department heads is a thoughtful addition — it helps explain why something was modified or declined.

Did you test submitting and tracking expenditure requests as a department head?

Client: Yes, I did, using one of the test accounts. It was easy to add expenditures to the spending table and then track their status from "Pending" to "Approved." The real-time updates are excellent — it makes communication between departments and HR so much smoother.

I'm glad to hear that. Let's move to the dashboards and data visualization. Were the graphs and tables clear?

Client: Absolutely. The dashboards are one of my favourite parts. The graphs showing planned versus actual spending make it easy to understand budget performance at a glance. I also liked that every role has a customized dashboard — department heads only see their figures, while admins can see everyone's overview.

If I could suggest one improvement, it would be to include a few more kinds of graphs — maybe a pie chart to show major expense categories, or monthly trends to analyse spending patterns better. That would make the comparisons between departments even more insightful.

That's great feedback. Did you find the budget report features helpful?

Client: Very much so. The reports are detailed and well-structured. I liked that I could generate them with a single click and receive them over email. The only thing is that the download is currently only in PDF format — it would be nice to have the option for Excel or Word files too, since some teams prefer to make their own notes in those.

Alright, noted. How did you find the notification system?

Client: That's another feature I really liked. Whenever a department submitted a budget, I immediately received an email notification. Similarly, department heads got notified when their requests were approved or rejected. It keeps everyone in the loop. The only small issue I noticed is that sometimes the emails take a few seconds longer to arrive, but it's not a big problem — just a minor delay.

Understood. How about the onboarding feature for new users?

Client: That's very helpful. When a new user account is created, they receive a simple guide explaining how to log in, what they can access, and how to use the dashboard. It

saves us time during training. I showed it to a few new employees, and they found it easy to follow.

I'm so glad that was useful. Was the system design and layout comfortable for regular use?

Client: Definitely. It's clean and well-organised. I liked the layout and how easy it is to navigate between sections. The only suggestion I'd make is to use slightly softer colours on the login and password pages — the bright background can be a little harsh. Maybe also increase the font size there a bit. But overall, the interface feels professional and user-friendly.

Thank you for mentioning that — I'll make a note of it. How did the overall performance feel, especially with multiple users working simultaneously?

Client: It worked smoothly! Even when several people were testing the system at the same time, there was no lag. I remember you telling me about using Node.js and MySQL for the backend — I'm not technical, but I think that's why it handled so many users efficiently. I've heard of both before, and I'm glad you used them because the system feels stable and fast.

That's great to hear. Are there any other small improvements you'd suggest?

C: Just a few little things:

- Maybe add a confirmation pop-up before deleting data, just to avoid accidental clicks.
- It would also be nice if the date and time of approvals appeared next to each budget request.

- And perhaps a simple “download all reports” button for admins, instead of doing it one by one.

These are all small details, but they’d make the experience even smoother.

Those are great points, thank you. Is there anything else you’d like to say about how the system has influenced your workflow?

Client: Overall, it’s made a huge difference. We no longer have to deal with messy Excel files or worry about data being lost. Everything is centralised, secure, and easy to monitor. It’s also saving us a lot of time during quarterly reviews. The transparency and accountability it brings are the biggest advantages.

Thank you so much, Mrs. [REDACTED] I really appreciate your detailed feedback and I’m glad the system has been useful for your team.

C: You’ve done an excellent job. It’s clear you really understood what we needed. I can definitely see this being used long-term to make our budgeting process smoother, faster, and much more reliable.

Client Signature : Oct 7, 2025.

Usha. R

AI Tool Used to transcribe interviews : TurboScribe